Part A: Sketching the Building Blocks

The single-variable functions every ML practitioner should be able to draw

Mohsin Raza · Fall 2026 **Due:** Tuesday 20 October 2026, before class

Why this matters. Many common machine learning models rely on simple functions that appear in activations, loss functions and probability calculations. If you can picture $\sigma(x)$, $\tanh x$ or $-\ln p$, you can reason about saturation, gradient behaviour and loss penalties without running code. These functions are also what you will differentiate in the chain rule and backpropagation.
How to work. Sketch *by hand first*, using the checklist below. Only then check yourself with a plot (see the last page). **Graded:** this is Part A of the Multivariable Calculus assignment, worth **25 of its 100 marks** (Part B, the multivariable part, is worth 75 and is released separately).
Show your reasoning. A1: justify each match with a distinguishing feature of the graph. A2–A3: show the calculations (derivatives, limits, critical points). A4: show the main algebraic or differentiation steps. Unsupported answers may receive reduced or no credit. Sketches are assessed on their mathematical features (asymptotes, intercepts, monotonicity, concavity, extrema), not on neatness. An answer key will be posted after the due date.
Submit: hand in your written work at the start of class on Tuesday 20 October. If you can't attend, upload a photo or scan to Google Classroom (Classwork → Module 5) by the same time.

The sketching checklist

For every function, work out these seven items, then draw.

1. **Domain and range.** Where is it defined? What values can it output?
2. **Intercepts and special values.** $f(0)$, zeros, any value worth knowing (e.g. $\ln 1 = 0$).
3. **Ends.** Limits as $x \to \pm\infty$ (or at the edge of the domain). Any asymptotes?
4. **First derivative** $f'(x)$. Where is f increasing ($f' > 0$) or decreasing ($f' < 0$)? Find the critical points *within the domain*, where $f'(x) = 0$ or $f'(x)$ does not exist, and decide whether each is a local maximum, a local minimum, or neither. Examine domain boundaries separately for possible extrema, even when the derivative is not zero there.
5. **Second derivative** $f''(x)$. Where is f concave up ($f'' > 0$, holds water) or concave down? Inflection points are where the concavity *changes*; look where $f''(x) = 0$ or does not exist, but note that $f'' = 0$ alone does not make an inflection point.
6. **Symmetry.** Even ($f(-x) = f(x)$), odd ($f(-x) = -f(x)$), or neither?
7. **Where it shows up in ML.** One sentence.

Worked example: the sigmoid $\sigma(x) = \dfrac{1}{1+e^{-x}}$

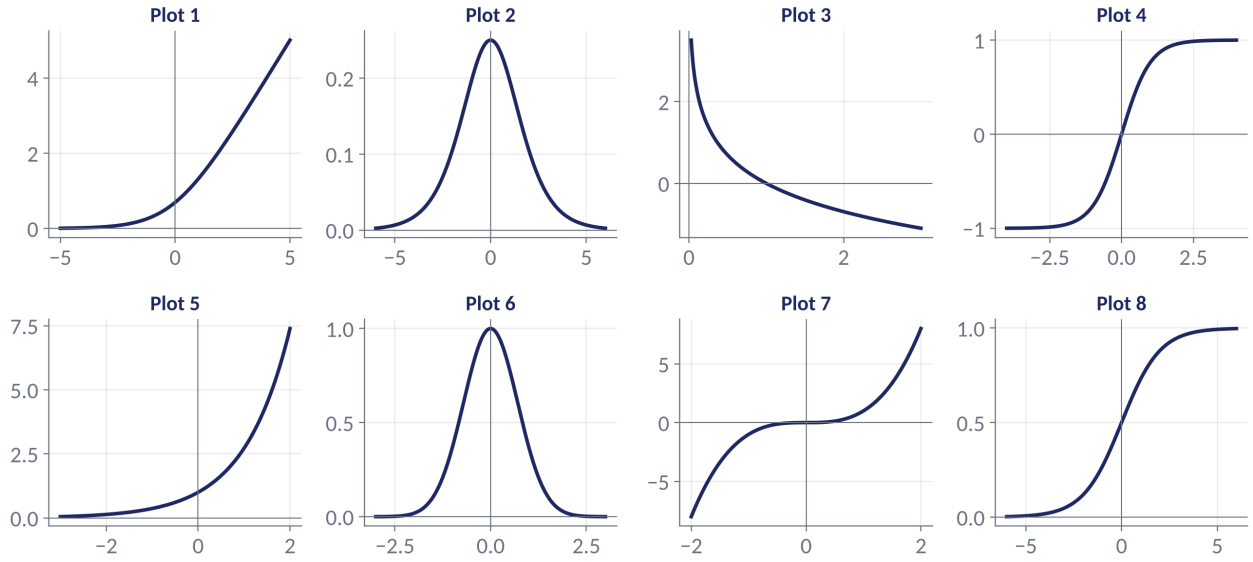
Domain / range	$\mathbb{R}$ / $(0, 1)$
Special values	$\sigma(0) = \frac{1}{2}$; no zeros
Ends	$\sigma \to 0$ as $x \to -\infty$, $\sigma \to 1$ as $x \to \infty$ (two horizontal asymptotes)
f'	$\sigma'(x) = \sigma(x)\,(1 - \sigma(x)) > 0$: always increasing, steepest at $x = 0$ with slope $\frac{1}{4}$
f''	$\sigma''(x) = \sigma'(x)\,(1 - 2\sigma(x))$: concave up for $x < 0$, down for $x > 0$; inflection at $(0, \frac{1}{2})$
Symmetry	neither, but $\sigma(-x) = 1 - \sigma(x)$ (point-symmetric about $(0, \frac{1}{2})$)
In ML	turns any score into a probability; logistic regression; classic activation

How to draw it:

1. Dashed asymptotes at $y = 0$ and $y = 1$.
2. Mark $(0, \frac{1}{2})$; draw a short line through it with slope $\frac{1}{4}$.
3. Left of 0: bend upward (concave up) from the $y = 0$ asymptote.
4. Right of 0: bend over (concave down) into $y = 1$.

A1. Recognize [8 marks]

Match each of the eight plots to its formula. Two formulas are not used. *Read the axis numbers:* some functions have similar shapes but differ in their intercepts, ranges, slopes or limiting behaviour.



(a) e^x (b) $\ln x$ (c) $-\ln x$ (d) $\sigma(x)$ (e) $\tanh x$
(f) $\mathrm{ReLU}(x)$ (g) $\ln(1 + e^x)$ (h) e^{-x^2} (i) $\sigma'(x)$ (j) x^3

For each plot, give the formula ($\frac{1}{2}$ mark) and one distinguishing mathematical feature that supports your choice ($\frac{1}{2}$ mark). "It is S-shaped" is not enough when more than one candidate has that shape: use something specific, such as the value at zero, the maximum height, the limiting values or the slope.

A2. The ML function zoo (do all four) [8 marks: 2 each]

For each function: the **full checklist, items 1–7**, then a labelled sketch by hand. Use the worked sigmoid example as a guide. Per function: $\frac{1}{2}$ mark for domain, range, intercepts, special values and limits; $\frac{1}{2}$ mark for derivatives, monotonicity, extrema, concavity and symmetry; 1 mark for an accurate, labelled sketch.

A2.1 e^x

A2.2 $\ln x$

A2.3 $\tanh x$

A2.4 $\mathrm{ReLU}(x) = \max(0, x)$. *At $x = 0$: what happens to the first and second derivatives? Is that point a corner, an inflection point, or both?*

A3. Beyond the zoo (choose any six) [6 marks: 1 each]

Choose any six of A3.1–A3.10. For each one: checklist **items 1–5** ($\frac{1}{2}$ mark), then a labelled sketch by hand showing the important features ($\frac{1}{2}$ mark). Items 6 and 7 are optional. Analyse restricted or disconnected domains interval by interval; wherever a derivative does not exist, say why.

	Function	Why it's worth knowing / what to watch for
A3.1	x^2	squared-error loss
A3.2	x^3	$f'(0) = 0$, yet it's not a maximum or a minimum. Why?
A3.3	$\lvert x \rvert$	absolute-error (L1) loss, Lasso; what happens to f' at 0?
A3.4	$1/x$	decreasing on each interval of its domain; is it decreasing over its *whole* domain? Compare $f(-1)$ and $f(1)$, and say why the distinction matters
A3.5	$\sqrt{x}$	standard deviations, RMS; slope as $x \to 0^+$? What does it say about the tangent at the origin?
A3.6	e^{-x^2}	the bell shape of a Gaussian, without its normalizing constant (Module 6); find its maximum and both inflection points
A3.7	$\text{softplus}(x) = \ln(1 + e^x)$	a smooth approximation to ReLU; find its horizontal asymptote as $x \to -\infty$ and its oblique asymptote as $x \to +\infty$
A3.8	leaky ReLU: x if $x > 0$, $0.1x$ if $x \le 0$	unlike ReLU, a nonzero slope on both sides of 0; is it differentiable at 0? Explain
A3.9	$-\ln p$, for $0 < p \le 1$	negative log-likelihood loss, with p the predicted probability of the *correct* class. What happens as $p \to 0$? Where is the minimum?
A3.10	$x \ln x$, for $x > 0$	a building block of entropy (Module 9); find its minimum and $\lim_{x \to 0^+} x \ln x$, and explain why the limit exists even though $\ln x \to -\infty$

A4. Connections [3 marks + 1 bonus]

A4.1 Transformations. [1] On one set of axes, sketch $\sigma(x)$, $\sigma(2x)$ and $\sigma(x - 3)$. For each, where is the midpoint ($y = \frac{1}{2}$) and what is the slope there?

A4.2 Two activations are one. [1] Show that $\tanh x = 2\,\sigma(2x) - 1$. *Hint: write the right-hand side as one fraction, then multiply top and bottom by e^x.* Explain in words how the sketch of σ turns into the sketch of tanh.

A4.3 Smooth ReLU, sigmoid slope. [1] Show that $\dfrac{d}{dx} \ln(1 + e^x) = \sigma(x)$. What does this say about the slope of softplus far to the left and far to the right?

A4.4 Bonus (+1): activation slopes and gradient flow. Using A4.2, show that $\tanh'(x) = 4\,\sigma'(2x)$, and hence $\tanh'(0) = 1$ and $\sigma'(0) = \frac{1}{4}$. Suppose a signal passes through several activations whose inputs are close to zero; during backpropagation, the gradient passing through each activation is multiplied by its derivative. Under these simplified conditions, which activation, sigmoid or tanh, is less likely to make the gradient shrink rapidly? Explain using their derivatives at zero. *(This compares slopes near zero only, not which activation is better in every network.)*

Check your sketches (after drawing by hand)

R

```
curve(plogis(x), -6, 6)   # sigmoid
curve(tanh(x), -4, 4)
curve(log1p(exp(x)), -5, 5) # softplus
curve(x * log(x), 0.001, 2)
```

Python

```
import numpy as np, matplotlib.pyplot as plt
x = np.linspace(-6, 6, 400)
plt.plot(x, 1/(1+np.exp(-x))); plt.show()
```

To check any function from A2–A3, change the formula in the same line, or use **desmos.com**. Where sketch and plot disagree, find the checklist item you got wrong. **Reading:** MML, Section 5.1.